M3D: MultiModal MultiDocument Fine-Grained Inconsistency Detection

Anonymous ACL submission

Abstract

Fact-checking claims is a highly laborious task that involves understanding how each factual assertion within the claim relates to a set of trusted source materials. Existing approaches make sample-level predictions but fail to identify the specific aspects of the claim that are troublesome and the specific evidence relied upon. In this paper, we introduce a method and new benchmark for this challenging task. Our method predicts the fine-grained logical relationship of each aspect of the claim from a set of multimodal documents, which include text, image(s), video(s), and audio(s). We also introduce a new benchmark (M^3DC) of claims requiring multimodal multidocument reasoning, which we construct using a novel claim synthesis technique. Experiments show that our approach significantly outperforms state-of-the-art baselines on this challenging task on two benchmarks while providing finer-grained predictions, explanations, and evidence.

1 Introduction

Misinformation poses serious societal risks by perpetuating narratives that incite fear, sow discord, and affect public health and safety (Geoghegan et al., 2020; Treen et al., 2020). Despite significant efforts towards developing automated fact-checking techniques (Yao et al., 2023a; Nasir et al., 2021; Karimi and Tang, 2019), existing methods face several limitations. First, real-world claims may include assertions that require consulting multiple documents and modalities to verify or refute the claim. Existing approaches either assume a single document setting (Fung et al., 2021; Thomas et al., 2022) or perform retrieval across documents to obtain relevant evidence, which is then treated as a single document (Yao et al., 2023a), potentially losing important surrounding context. Secondly, some methods only predict when claims conflict with relevant knowledge but ignore ambiguous cases where no supporting or refuting information is available (Wu et al., 2022; Xuan et al., 2024). Lastly, most of the existing methods fail to provide the fine-grained analysis needed for users to understand what is inconsistent in a claim or to make revisions to be more factual (Wu et al., 2022; Yao et al., 2023a; Xuan et al., 2024). Simply flagging an entire claim as false without pinpointing the specific inaccurate parts provides limited utility.

In contrast, we propose an approach for predicting the logical relationship of each piece of a claim with respect to a set of multimodal sources. We perform a semantic dissection of claims into semantic pieces and leverage a hierarchical transformer that operates across multimedia documents to make fine-grained predictions. Our model ingests the claim along with associated multimedia, preserving the context. It then fuses the cross-document representations into a graph initialized with the claim's Abstract Meaning Representation (AMR) (Banarescu et al., 2013). Entailment relations are then predicted for each node (e.g., entities, actions) and tuple (e.g., relations) within the graph.

Because no prior work has explored making fine-grained claim predictions from a set of multimodal documents, we also introduce a new dataset of claims that contains fine-grained labels for this task called M^3DC (**M**ulti**M**odal **M**ulti-**D**ocument **C**laims). We build our dataset on top of the NewsStories(Tan et al., 2022a) dataset, which includes sets of news articles, images, and videos across multiple topics. We retrieve textual, visual, and audio data from each set to build a robust multimodal multidocument knowledge graph for each set of related documents. Next, we develop a claim synthesis method in order to generate claims that require multisource knowledge to verify, which uses a fine-grained claim manipulator model to generate claims manipulated at the sub-claim level.

Our major contributions are as follows:

- We introduce the novel task of performing

fine-grained entailment of a textual claim with a set of multimodal documents.
- We introduce a new hierarchical transformer model designed for the task of fine-grained claim analysis over multiple sources.
- We propose a novel data synthesis technique for generating fine-grained labeled claims requiring multimodal multisource knowledge to verify using a graph traversal and fine-grained claim manipulator model.
- We contribute a large benchmark of fine-grained labeled claims created using our technique. We also contribute a small number of claims densely annotated by experts.
- We conduct qualitative and quantitative experiments to evaluate the performance of our proposed method on our new benchmark dataset, as well as an existing benchmark dataset.

2 Related Works

2.1 Fact-checking Datasets

Fact-checking claims is an established task with many text-only benchmarks such as LIAR (Wang, 2017), SNLI (Bowman et al., 2015), MNLI (Williams et al., 2018), and ANLI (Nie et al., 2020a). Later datasets like FEVER (Thorne et al., 2018) and SciFact (Wadden et al., 2020) incorporate evidence from multiple text-only sources. Recent work has explored fine-grained, text-only misinformation detection across multiple sources, such as (Wu et al., 2022), but ignores that crucial evidence can also come from other modalities.

To address this, datasets like MM-Claims (Cheema et al., 2022a) and Fakeddit (Nakamura et al., 2019) have incorporated multimodal fact-checking data. However, these tend to assess authenticity at a coarse level, with claims derived from short sentences or paragraphs. In contrast, our proposed dataset features extremely fine-grained labels that capture the logical relationships between each aspect of a claim and a set of multimedia documents. This requires a much more nuanced entailment analysis and a deeper understanding of the evidence required to verify or refute specific claim components. Furthermore, our data synthesis process explicitly creates claims from multiple modalities across various documents with lengthy sentences. Our approach explicitly models these intricate multimodal relationships and aims to facilitate the development of more interpretable and trustworthy fact-checking systems.

2.2 Fake News Detection Methods

Many methods for detecting misinformation at the document level have been proposed (Zellers et al., 2019; Atanasova et al., 2019; Karimi and Tang, 2019). These approaches either directly rely on neural models (Nasir et al., 2021) or rely on structural cues indicative of disinformation (Karimi and Tang, 2019). Hu et al. (2021) compares documents to an external knowledge graph (KG) for verification, but it does not make fine-grained predictions or leverage multimodal data. Predicting claim entailment from evidence has been studied in NLP (Giampiccolo et al., 2007; Lai and Hockenmaier, 2017; Han et al., 2017) and computer vision (Kayser et al., 2021; Yang et al., 2022), using visual instead of textual evidence.

Recent multimodal misinformation detection approaches (Yao et al., 2023a; Tan et al., 2020; Singh et al., 2021) make entailment predictions from multimodal documents. MOCHEG (Yao et al., 2023a) predicts entailment at the sample level. Fung et al. (2021) extracts a multimodal KG for fake news detection, focusing on internal inconsistencies in a single document. Wu et al. (2022) propose a GNN-based model for fine-grained predictions across text-only documents, using IE-generated KGs which may miss fine details. Unlike prior work, our proposed method performs fine-grained entailment of complex claims against a set of multimodal documents. Wu et al. (2022) can only output binary inconsistency results using limited IE-generated knowledge graphs. Thomas et al. (2022) considers shorter, caption-like claims verifiable from a single image. In contrast, our claims are much more complex, requiring reasoning over a multimodal, multi-document premise. Unlike Thomas et al. (2022), we further contribute a new benchmark dataset for this task.

3 Approach

In this section, we first describe our methodology for constructing our dataset, M^3DC. We then provide details of our model architecture which operates across sets of documents to make fine-grained claim predictions.

3.1 Multimodal MultiDocument Dataset

In this section, we introduce our data synthesis approach for constructing a dataset with claims containing fine-grained labels that require multimodal and multi-source knowledge to verify. Our dataset

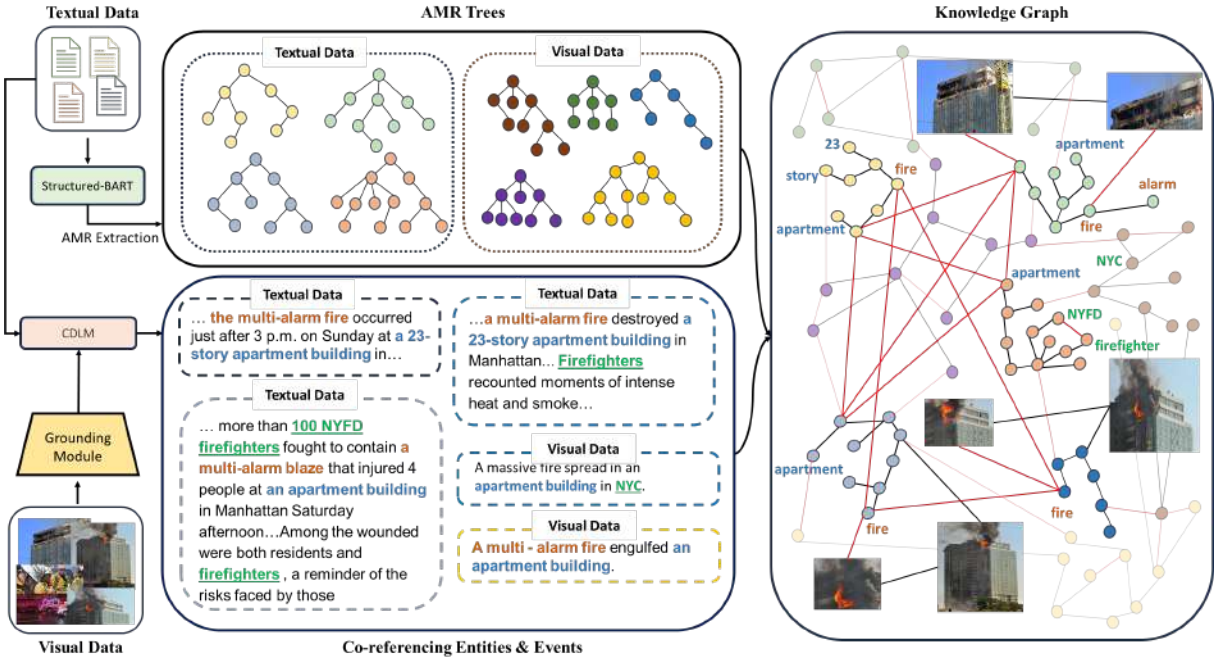


Figure 1: Constructing a KG from a multimedia news cluster. AMR trees from different documents and modalities are linked to form a cross-document, cross-media KG. Co-reference links are shown in red.

builds upon NewsStories (Tan et al., 2022b), a collection of news clusters with articles and videos. We begin by crawling the data and removing news that is no longer publicly accessible or has been taken down. For each news cluster, we construct a knowledge graph (KG) combining textual and non-textual data based on AMR trees (Banarescu et al., 2013) generated from news documents. This cross-document, cross-media representation allows us to synthesize claims by linking information from the graph. We then introduce a claim manipulator model that generates claims with varying degrees of truthfulness by traversing the AMR-based KG and introducing controlled perturbations. To obtain fine-grained labels, we employ a model that assigns entailment labels (e.g., entailment, contradiction, neutral) to individual AMR nodes and tuples with its associated knowledge. Using this approach, we synthesize a dataset of about 400K claims across over 10,000 topics, requiring multimodal and multi-document knowledge for verification. The overall process is shown in Figure 2.

3.1.1 Knowledge Graph Construction

For each news cluster, we extract knowledge into a set of AMR trees using Structured-BART(Zhou et al., 2021) with sentences coming from the news document, visual captions generated from our grounding module and audio summaries from Qwen-Audio(Chu et al., 2023). Then, we connect nodes from AMR trees using co-reference resolution from CDLM(Caciularu et al., 2021) and F-coref (Otmazgin et al., 2022) in order to link within-document and cross-document entities or events. The overall process is illustrated in Figure 1.

For visual data from images and videos, we utilize GLIP (Li et al., 2022) and CoFormer (Cho et al., 2022) to perform co-reference resolution across modalities. Initially, CoFormer is used to extract event frames from each image and subsampled video frames, along with grounded bounding boxes. Event frames are defined by PropBank (Palmer et al., 2005) and contain event verbs, along with noun arguments and roles. Meanwhile, GLIP (Li et al., 2022) is used to ground textual data from news articles with visual content from images or video frames. After grounding textual and visual content using both models, we measure the Intersection over Union (IoU) between groundings from both models to filter out discrepancies. Then we utilize GPT-Neo (Black et al., 2021) to generate captions for each image and video frame from the event frame extracted by CoFormer and textual data grounded from GLIP, using an in-context learning approach with a pre-defined template. For audio data, we prompt Qwen-Audio (Chu et al., 2023) to generate summaries that describe the audio content and background noise from the video.

3.1.2 Claim Generation

To generate claims that require multimodal, multi-document evidence from the constructed KGs, we developed a Depth-First Search (DFS) based graph traversal method that selects Knowledge Elements (KEs) from multiple sources from the constructed KG. For a given KG and starting node (i.e. an AMR

predicate node), the traversal algorithm traverses surrounding nodes until another predicate node is reached. We encourage the algorithm to follow coreference edges to incorporate knowledge across documents and modalities. The traversal algorithm outputs KEs (AMR triples) rooted at a predicate, which is then used to generate a complete claim sentence containing the information from the traversed nodes and edges through AMRBART (Bai et al., 2022). Given that these generated claims are directly generated from the KG, all resulting claims are inherently entailed by this approach. This approach ensures that the resulting claims rephrase evidence from different articles and modalities, requiring the model to reason across sources to perform fine-grained verification.

3.1.3 Claim Manipulation

Since the claims generated directly from the KGs are inherently entailed, we introduce a claim manipulator model to generate diverse claims with varying degrees of truthfulness (entailed, neutral, or contradicted) relative to the evidence in the KG. The claim manipulator takes as input the claim, relevant evidence from the KG (which may be multimodal), and a desired logical label (entailed, neutral, or contradicted). The goal is to manipulate an entailed claim so that the claim's logical relation matches the input. To train the manipulator, we employ reinforcement learning, where a model is optimized to maximize the scores provided by a reward model that offers evaluative feedback.

Denoting the original claim as c, derived from the KG, and the modified claim as $\hat{c}$ produced by the manipulator M, with y representing the logical label from $\mathbb{Y} = \{"entailed", "neutral", "contradicted"\}$, the goal of the claim manipulator is to generate a claim similar to the original claim c with the target logical label $\hat{y}$ given premise (evidence) p. We leverage Llama-2-13B (Touvron et al., 2023) to manipulate claims to correspond with the designated logical label $\hat{y}$ based on the given premise p. The premise consists of the top 10 most relevant evidence (expressed in text, i.e., using sentences from news articles and captions for image and video) related to c from Sentence-BERT(Reimers and Gurevych, 2019), the manipulator is fine-tuned using reinforcement learning to produce a claim $\hat{c}$ based on c. In this process, c and $\hat{c}$ are intended to be syntactically similar to each other. The claim manipulator can be formulated as $\hat{c} = M_\theta(p, c, \hat{y})$

To steer the manipulator towards generating claims that align with the target logical label $\hat{y}$ and similar to the original claim c syntactically, a reward model based on DeBERTAv3 (He et al., 2023) is trained to function as a critic using MNLI (Williams et al., 2018), Fever-NLI (Thorne et al., 2018), and ANLI (Nie et al., 2020b). The reward model is trained for fine-grained entailment classification using the multi-instance and structural constraints from FGVE (Thomas et al., 2022), allowing the model to do fine-grained predictions without ground-truth fine-grained labels. Critically, we enforce our target label constraint at both the fine-grained and sample levels within the graph. This approach ensures that the claim manipulator not only focuses on producing claims in a coarse-grained manner but also pays attention to fine-grained details. Specifically, the reward model's score is defined as the likelihood of the target label considering both the manipulated claim and the top 10 sentences most relevant to the original claim from the KG (serving as evidence):

$$r(c, \hat{c}, \hat{y}) = P(\hat{y} \mid p, \hat{c}) - \left(\sum_{y_i \neq \hat{y}}^{|\mathbb{Y}|} P(y_i \mid p, \hat{c}) + \text{ROUGE}(c, \hat{c})\right) \quad (1)$$

where c, $\hat{c}$, $\hat{y}$, and p represent the original claim, the modified claim, the desired logical label for the claim, and the premise, respectively. The term $P(\hat{y} \mid p, \hat{c})$ is obtained from the trained fine-grained entailment classifier. The goal of this reward function is to ensure that the modified claim $\hat{c}$ not only matches the intended truthfulness label $\hat{y}$ but also retains as much similarity to the original claim c as possible as quantified by the ROUGE score.

We fine-tuned the claim manipulator with Proximal Policy Optimization (PPO) (Schulman et al., 2017) as our policy gradient method for reinforcement learning. PPO adds an additional term to the reward function, which imposes a penalty determined by the Kullback-Leibler (KL) divergence between the trained RL policy manipulator, π_ϕ^{PPO}, and the initial supervised manipulator π^{SFT}:

$$r_{total} = r(\hat{h}, h, \hat{y}) - \eta KL(\pi_\phi^{PPO}(\hat{y}_t \mid p, \hat{h}), \pi^{SFT}(\hat{y}_t \mid p, \hat{h})), \quad (2)$$

where η represents the KL reward coefficient, which determines the magnitude of the KL penalty; we set it to 0.2 for our model. This coefficient functions as an entropy boost, enhancing exploration throughout the policy domain and urging the model

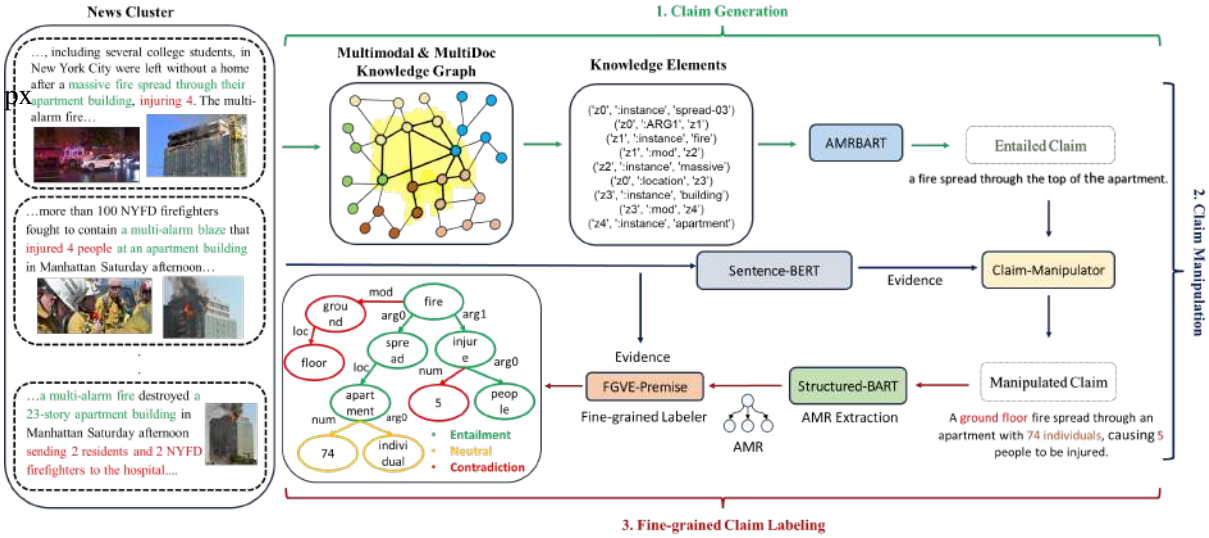


Figure 2: Claim generation pipeline. We create a knowledge graph from a set of media about an event. Our traversal algorithm selects the part of the KG highlighted in yellow to generate a (true) claim. To do so, we use the selected elements to translate the selected knowledge into a sentence. We then feed relevant evidence and the generated claim into our claim manipulator model. In this example, we ask our claim manipulator to generate a contradicted claim. The claim manipulator performs fine-grained manipulations, inserting both unverified (i.e. 74 individuals) and contradictory (i.e. 5 people injured) assertions. Because we know how the claim was manipulated at the knowledge-element level, we can use this as supervision to train our verification model.

to engage in a diverse set of actions rather than the one currently considered the best. In addition, it inhibits the policy from rapidly committing to a singular strategy, and this encourages outputs from the RL fine-tuned model to not deviate too far from the original model. After constructing the dataset with the claim manipulator, we employ Mixtral-8x7B (Jiang et al., 2024) using in-context learning to predict the logical label of the claims generated by the claim manipulator as a quality check; we discard those that do not align with the target labels. Finally, as a final quality check on our generated dataset, we assess the checkworthiness of claims using ClaimBuster (Arslan et al., 2020) to filter opinions or unimportant claims from our dataset. More details are covered in Appendix A.1.

3.2 Model Architecture

In this section, we present our model for predicting fine-grained entailment relations for claims given a set of trusted multimodal source materials. Figure 3 shows our model's architecture.

3.2.1 Multimodal Encoder

By design, our claims require reasoning across modalities and documents to make fine-grained predictions. We thus integrate all modalities into our model, preserving the original context in which the claim appeared. For textual content, we employ LongT5 (Guo et al., 2022) to encode the claims and sentences from documents and captions. For handling non-textual context (i.e. images, video, and audio), we utilize ImageBind (Girdhar et al., 2023), a set of cross-modal embedding models for embedding text, audio (represented as spectrograms), visual content, and other modalities in a common space. In addition to explicitly capturing how the information relates across documents and modalities, our model also ingests an embedding of the KG corresponding to each cluster. To learn our KG embedding, we instantiate our KG using a Graph Convolutional Network (GCN) and train it via a masked sequence prediction task. We randomly obscure nodes and edges within the KG and train a classifier to predict the masked pieces. After training, we extract KG embeddings for each cluster and feed them to our model. To bridge the various representation spaces, we add an additional linear layer for each modality's encoder.

The embeddings from different modalities, including textual content (encoded by LongT5), non-textual context (encoded by ImageBind), and the knowledge graph (encoded by the GNN), are concatenated to form a comprehensive multimodal representation of the claim and its associated evidence. This concatenated embedding is then fed into LongT5(Guo et al., 2022) for pretraining using the Gap Sentence Generation (GSG) objective from Pegasus (Zhang et al., 2020). GSG is a self-supervised learning task that aims to generate missing sentences within a given context. We identify the top 3 sentences inside the news documents that

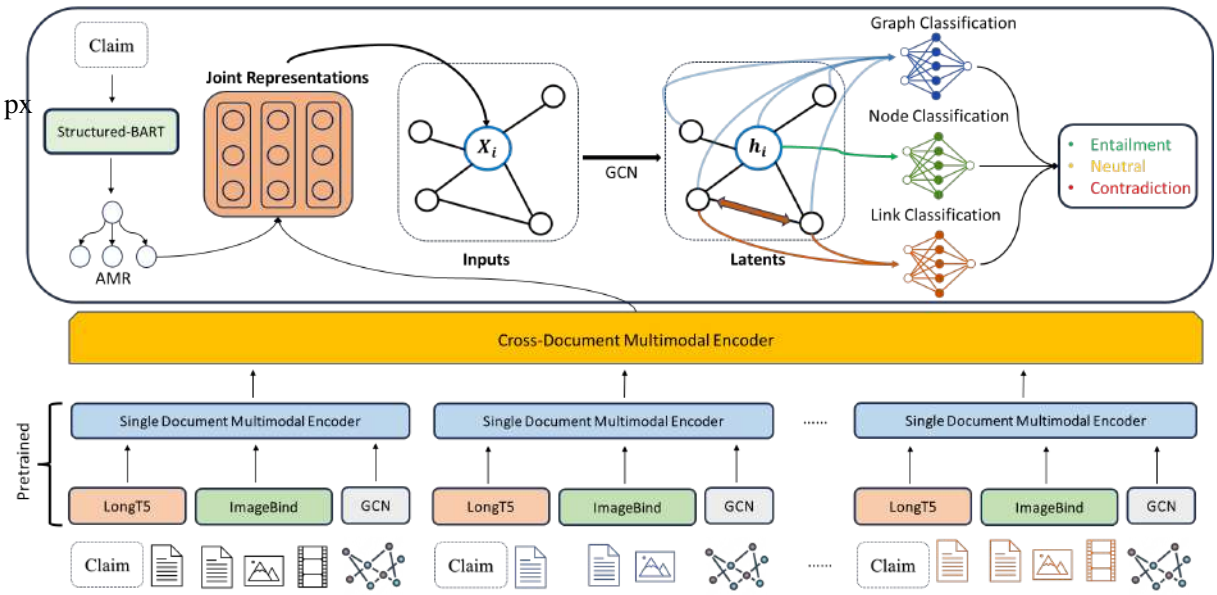


Figure 3: The model architecture. Each cluster, potentially containing multiple news articles, will have its content from various multimedia sources independently encoded and then merged to form a unified representation. This joint representation will serve as the initial state for every node within the GNN. Subsequently, labels at both the sample level and the fine-grained level can be derived by aggregating features from the nodes and edges of the GNN.

are most relevant to the claim c using ROUGE-F1, randomly choose one sentence and its adjacent sentence, and then mask them both. LongT5 is trained to generate the masked sentences based on the surrounding context and the multimodal embeddings.

3.2.2 Graph Convolutional Network

Our task requires predicting fine-grained entailment relationships between a claim and a set of multimedia source materials. To ensure each fine-grained element within the claim's AMR captures the context of the AMR structure in which it appears, we employ a two-layer GCN (Kipf and Welling, 2016) to learn contextual features of each node and tuple within the claim's AMR graph. Our GCN model is initialized with features from our multimodal encoder and features from the claims's AMR. Specifically, we encode the AMR representation of claims and embeddings from multimedia news content via the GCN as follows: for each node i within the AMR graph, we define the feature aggregation mechanism by the equation:

$$h_i^{(l+1)} = \sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{c_{ij}} h_j^{(l)} \quad (3)$$

where $h_i^{(l+1)}$ is the feature vector of node i at the subsequent layer $l+1$. The set $\mathcal{N}(i)$ includes the neighbors of node i, and c_{ij} is a normalization factor for the edge that connects nodes i and j.

For edge features, we extend our model to incorporate edge features alongside node features. This is achieved by incorporating edge attributes into the aggregation function, allowing the model to consider the characteristics of the connections between nodes. For an edge e_{ij} connecting nodes i and j, the edge features can be integrated as follows:

$$e_{ij}^{(l+1)} = \left[W_e^{(l)} h_i^{(l)} || W_e^{(l)} h_j^{(l)} \right] \quad (4)$$

where $e_{ij}^{(l+1)}$ represents the feature vector of edge e_{ij} at layer $l+1$, with $W_e^{(l)}$ and $b_e^{(l)}$ being the weight matrix and bias vector specific to edge features at layer l. This approach ensures that the model captures not only the node-level but also the edge-level semantic and structural information inherent in AMR graphs.

For graph-level (sample-level) classification, we aggregate the features of the entire graph with average pooling. Finally, multiple MLP classifiers are applied to make predictions for nodes, edges, and the graph on the sample-level and fine-grained tasks. We train our model using cross-entropy loss with labels from the trained fine-grained entailment classifier in section 3.1.3.

4 Experiments

4.1 Multimodal MultiDocument Dataset

We compare our new dataset with others in Table 1. Our dataset contains fine-grained labels across 180,000 entailed claims, 121,224 neutral claims, and 113,181 contradicted claims. While existing datasets are topic-specific, our claims are highly detailed and topically diverse. In our supplementary, we include examples of claims from our dataset compared to other datasets.

Datasets	#Samples	Data source	Topic(s)	Multi-modality	Multi-document	Claim verification	Fine-grained Labels
Zlatkova et al. (2019)	1,233	Snopes, Reuters	Multi	✔	✗	✔	✗
Cheema et al. (2022b)	3,400	X	COVID-19, Climate change, Technology	✔	✔	✔	✗
Nielsen and McConville (2022)	12,914	X	Multi	✔	✔	✔	✗
Yao et al. (2023b)	15,601	Politifact, Snopes	Multi	✔	✗	✔	✗
Nakov et al. (2021)	18,014	X	COVID-19, Politics	✗	✔	✗	✗
Ours	414,405	Multi-News	Multi	✔	✔	✔	✔

Table 1: Comparison between different datasets in terms of multi-modality, multi documents, claim verification, and fine-grained labels. Ours is the largest one that supports fine-grained labels with multimodal document claim verification. No dataset provides fine-grained labels.

Model	Synthetic Labels								Human Labels							
	Sample-level				*Fine-grained*				*Sample-level*				*Fine-grained*			
	E	N	C	All	E	N	C	All	E	N	C	All	E	N	C	All
FGVE (Thomas et al., 2022)	0.27	0.2	0.28	0.25	0.23	0.1	0.09	0.14	0.32	0.14	0.36	0.27	0.30	0.05	0.04	0.13
MOCHEG (Yao et al., 2023a)	0.32	0.14	0.36	0.27	0.28	0.13	0.32	0.24	0.37	0.18	0.41	0.32	0.35	**0.14**	**0.39**	0.29
LLaVA-1.5 (Liu et al., 2023)	0.57	0.0	0.33	0.30	**0.73**	0.0	0.14	0.29	0.67	0.0	0.43	0.37	**0.88**	0.0	0.13	0.33
MiniGPT-v2 (Chen et al., 2023)	0.50	0.0	0.43	0.31	0.56	0.0	0.24	0.27	0.62	0.0	**0.62**	0.41	0.54	0.0	0.09	0.21
Ours	**0.72**	**0.26**	**0.48**	**0.49**	0.65	**0.23**	**0.41**	**0.43**	**0.72**	**0.21**	0.59	**0.51**	0.68	0.1	**0.39**	**0.39**

Table 2: Results on our M^3DC benchmark. We report class-wise F1 scores (E: entailed, N: neutral, C: contradicted) and the overall F1 score (All).

4.2 Testing Datasets and Baselines

We evaluate entailment performance on our M^3DC benchmark and MOCHEG (Yao et al., 2023a). We report F1 scores for each of the three classes and the macro-averaged overall F1. We evaluate both sample and fine-grained level prediction tasks. For M^3DC, we compare using two types of labels: human annotated labels and synthetic labels generated from the fine-grained entailment classifier (Section 3.1.3). For human labels in M^3DC and fine-grained labels in MOCHEG, we randomly selected 30 samples for human annotation using six expert annotators. Performing human annotation on samples is an extremely laborious and time-consuming task, which requires annotators to read multiple documents, look at multiple images, and watch video(s), which are often quite long. Annotators then must densely annotate every KE within the claim. We include additional details on our human study in our supplementary. Our baselines consist of a set of large vision-language models (LVLM) that have demonstrated strong performance on various multimodal tasks. Task-specific baselines include FGVE (Thomas et al., 2022) and MOCHEG (Yao et al., 2023a), while LVLM baselines include LLaVA-1.5 (Liu et al., 2023) and MiniGPT-v2 (Chen et al., 2023).

4.3 Quantitative Evaluation

Table 2 shows our model outperforming baselines on the M^3DC dataset, with similar results on synthetic and human-labeled data. This is critical, as it shows that the performance of our models on our human-annotated data tracks closely with the performance obtained on our large synthetic dataset, suggesting our synthetic dataset is a good evaluation benchmark for this task. On the MOCHEG dataset (Table 3), our model outperforms in fine-grained predictions, despite being trained on a diverse news dataset, M^3DC, rather than MOCHEG. While LLaVA and MiniGPT-v2 outperform in overall F1, they fail to correctly identify neutral claims, which our model handles better. The MOCHEG dataset's lack of video and audio and different styles of text (Snopes vs News) contributes to its lower performance at the sample level.

Model	*Sample-level*				*Fine-grained*			
	E	N	C	All	E	N	C	All
FGVE (Thomas et al., 2022)	0.37	0.16	0.37	0.3	0.31	0.1	0.2	0.20
MOCHEG† (Yao et al., 2023a)	0.57	0.23	0.40	0.39	0.52	**0.21**	**0.36**	0.37
LLaVA-1.5 (Liu et al., 2023)	0.67	0.0	**0.93**	**0.53**	0.44	0.0	0.25	0.23
MiniGPT-v2 (Chen et al., 2023)	0.67	0.0	**0.93**	**0.53**	**0.71**	0.0	0.25	0.32
Ours	**0.69**	**0.25**	0.48	0.47	0.63	0.18	**0.36**	**0.39**

Table 3: Results on MOCHEG dataset (Yao et al., 2023a). *All labels are human labels in this benchmark.* We report class-wise F1 scores (E: entailed, N: neutral, C: contradicted) and the overall F1 score (All). †: Note that MOCHEG (Yao et al., 2023a) is also trained on this dataset, while *our method is applied zero-shot*.

Model	*Sample-level*				*Fine-grained*			
	E	N	C	All	E	N	C	All
Ours w/ Text	0.69	0.25	0.43	0.46	0.61	0.15	0.34	0.37
Ours w/ Text + Image	0.71	**0.26**	0.42	0.46	0.63	0.18	0.36	0.39
Ours w/ Text + Image + Video	**0.72**	**0.26**	**0.48**	**0.49**	**0.65**	**0.23**	**0.41**	**0.43**
Ours w/ Text + Image + Video + Audio	0.70	0.24	0.47	0.47	0.63	0.21	**0.41**	0.42
Ours All w/o Text	0.42	0.02	0.29	0.24	0.37	0.01	0.23	0.20

Table 4: Ablation on M^3DC showing the impact of removing different modalities on our method.

4.4 Ablations

To demonstrate our model's capability in handling multimodal inputs, we conducted ablation studies

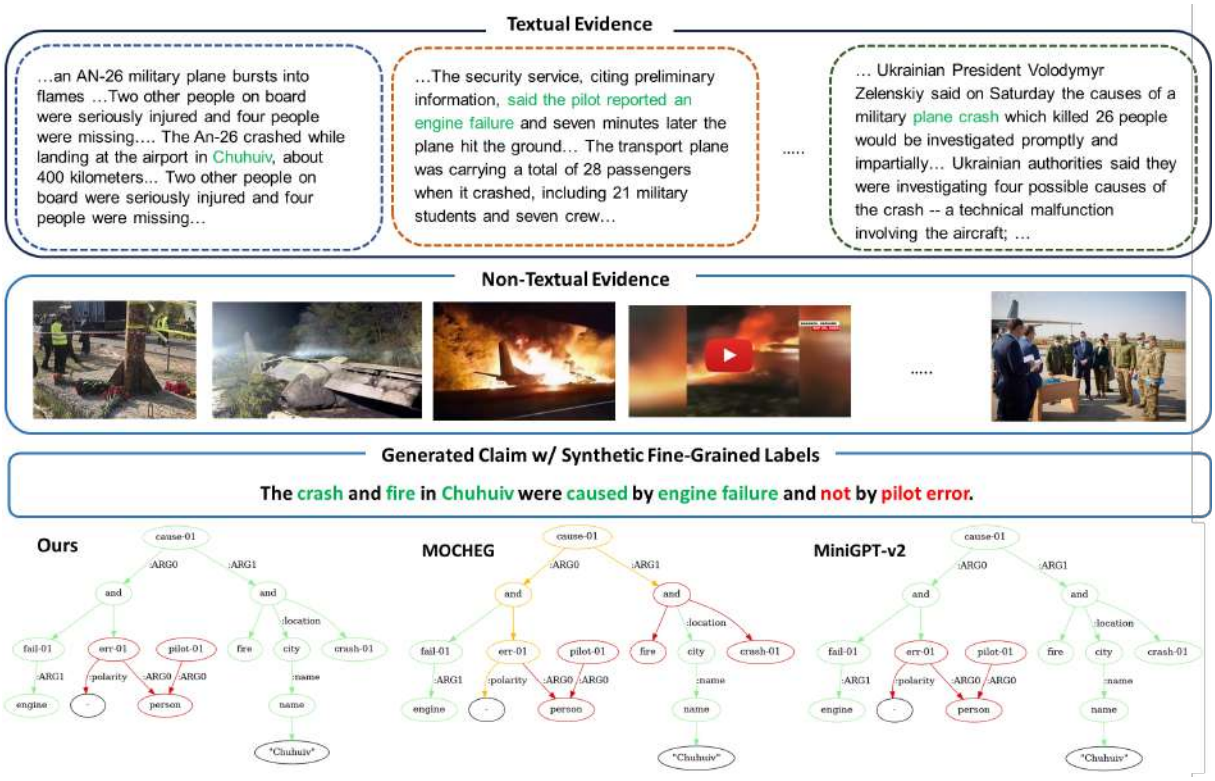


Figure 4: Qualitative results comparing our method's fine-grained predictions with those obtained from other baselines. We include additional results in our supplementary materials.

with varying combinations of modalities, as outlined in Table 4. Considering that a substantial portion of the information in KGs is derived from the textual content of news articles, it was anticipated that the text modality would play a pivotal role in the model's inference process. Our results, however, indicate that including additional modalities, such as visual and audio, did not significantly enhance the model's performance. This observation suggests that the dominance of text-based claims in our dataset may lead the model to prioritize textual features, which are typically sufficient for classifying claims derived from textual information.

4.5 Qualitative Results

We show qualitative results comparing our method with competitive baselines in Figure 4. We illustrate predictions on nodes and tuples by the color of the edges (green=entailed, yellow=neutral, red=contradiction). Node colors indicate node predictions, while edge colors represent tuple predictions. We perform fine-grained claim verification for the claim "The crash and fire in Chuhuiv were caused by engine failure and not by pilot errors." In actuality, the crash was partially caused by pilot errors, so this portion of the claim is shown in red (as being contradicted by certain media sources). We observe that our method identifies the correct portion of the claim as being contradicted by the evidence, while baselines tend to make more random predictions throughout the graph. In general, we observe that our GNN-based approach is able to ensure more semantic consistency across predictions than approaches that make predictions for each part of the claim separately.

5 Conclusion

In this paper, we address the challenge of predicting the logical consistency of claims with multimodal sources. Our method analyzes claims within a multimodal multidocument context, including text, visual content, and audio. Our method is able to reason in a fine-grained manner over complex information across media and modalities. We further introduce a dataset, M^3DC, created through a unique synthesis technique that produces claims requiring cross-document, cross-media reasoning for verification. This benchmark will enable the evaluation of multimodal fact-checking models and spur further research in this space. Our contributions aim to mitigate the impact of misinformation and enhance the reliability of automated fact-checking systems, thus supporting informed decision-making and fostering a factually accurate public dialogue.

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A Appendix

A.1 Dataset Analytics

In this section, we present additional details about our dataset, M^3DC. To demonstrate that the claims in our dataset do not rely solely on textual data, we provide examples in Figures 5 and 6 that incorporate information from images and videos. Figure 5 showcases claims generated from image evidence selected from the KG and claims derived from knowledge elements that co-reference the visual content. This approach ensures that the generated claims contain a degree of visual information. These claims are then modified by the claim manipulator to integrate data from different modalities and documents. As a result, the final claims not only reflect the representative visual content but also potentially include the underlying context behind the image or related information from the news articles. By incorporating visual evidence and manipulating claims to integrate multi-modal data, our dataset presents a diverse set of claims that require both textual and visual understanding for verification. This highlights the importance of considering information from various modalities when assessing the veracity of claims in real-world scenarios.

A.2 Qualitative Results

To provide insight into the dataset and the results from our model, we provide additional examples. From Figure 7, 8 and 9, we show a random selection of the M^3DC dataset and the results from our model, respectively. According to the results shown in the figures, it is evident that the majority of the generated claims require detailed evidence to be properly reasoned about. Furthermore, the results demonstrate that our model is able to accurately reason about these claims, as most of the model's outputs are correct when compared to the evidence provided by the documents. This suggests that our model is capable of effectively utilizing the available evidence to make accurate predictions, even when the claims are complex and require careful consideration of multiple pieces of information.

Despite the promising results, it is important to note that the majority of the generated claims do not rely heavily on visual data. This can be attributed to the nature of news articles, where most of the information is conveyed through textual content, and visual data may not provide a significant amount of additional evidence, as shown in the provided examples. Consequently, the performance of our model on this dataset may not fully showcase its ability to reason about claims that are more visually-centric. To address this limitation and further evaluate the capabilities of our model, future studies could explore its performance on datasets that place a greater emphasis on visual information, such as the Flicker dataset. By testing our model on a more visually centric dataset.

A.3 Human Annotations and Statistics

A.3.1 Annotation Details

In Table 2, we investigate the results of our model on human-labeled data to evaluate the performance of human annotators. To measure the inter-annotator agreement (IAA), we employ two annotators for each news cluster, with thirty different news clusters in total, who are responsible for labeling both the sample-level and fine-grained labels.

The inter-annotator agreement can be defined using the following formula:

$$IAA = \frac{\text{Number of samples with matching labels}}{\text{number of samples}} \quad (5)$$

Although our human-labeled dataset contains only 30 samples, annotating each claim derived from a news cluster can be a time-consuming process, taking anywhere from 30 to 60 minutes, depending on the number of news documents in each cluster. This is because many of the claims in our dataset rely on small details scattered across multiple news documents to determine the logical label at the sample level, which can be challenging even at the fine-grained level.

Our annotation interface power by Label Studio is shown in Figures 10, 11, and 12. For each news cluster, the annotators are required to go through a series of documents with multiple images and videos to determine the logical label of the claim. As shown in Figure 10, our interface displays the textual and image content of the news cluster, where each cluster could contain up to five different news documents. In addition to the textual and image content, each news document could be linked to one or more corresponding videos, as shown in Figure 11. The annotators are required to review every video as well, and the number of videos could sometimes be up to a dozen. After reviewing all the available information, the annotators need to label the sample-level label first according to the given claim. For each AMR tuple, the



The image depicts a United Airlines passenger jet flying through a cloudy sky, but the aircraft is actually a Boom Supersonic jet, not a United Airlines plane.

The image does not feature a sleek and modern airplane, but rather an outdated and cumbersome aircraft that lacks the advanced technology and design of contemporary airplanes.



The closure of the hotel was inevitable due to the preventable death of a young man and the dangerous conditions that were not addressed by the owners.

The police officers in the image are not responding to an incident or patrolling the area, but are instead posing for a photo shoot to promote the hotel's new crime-fighting initiative.



Despite the lack of dedicated state-led efforts, the Louise Michel's mission to save lives at sea should not be penalized or stigmatized, as it fills a critical gap in the absence of effective repatriation agreements.

The Italian government's decision to not take in more migrants has led to a continued absence of dedicated EU-led search and rescue capacity in the Central Mediterranean, resulting in a surge of migrant arrivals in Italy this year.



The recent explosion of SpaceX's Starship rocket during a landing attempt highlights the challenges and risks associated with the development of advanced space technology.

The rocket launch in the image is not as successful as it appears, as the rocket is surrounded by smoke and the sky is cloudy, indicating a potential malfunction or setback.



The recent explosion in Nashville has caused significant damage to the city's infrastructure, including the AT&T facility, which has resulted in disruptions to service and flight restrictions in the area.

The image depicts a thriving city street with modern buildings and well-maintained infrastructure, contradicting the notion of urban decay and destruction.

Despite the recent explosion in Nashville, the city's infrastructure remains intact, with the exception of a few damaged buildings and streets.



The man in the boat is experiencing a challenging journey through the flooded area due to the severe weather conditions.

The man in the boat is taking a risky journey through the flooded area, as the closed roads and severe weather conditions may make it difficult for him to reach his destination safely.

The man in the boat is not navigating through a flooded area, but rather through a serene and peaceful lake, with the surrounding trees providing a picturesque background.

Figure 5: Claims generated by our pipeline, with entailed, neutral, and contradicted claims denoted by green, orange, and red dashed lines, respectively. Claims based on image content are generated by selecting knowledge elements rooted in the image nodes of the KG. Then the claim manipulator adjusts the claim based on the premise, allowing control over the degree of evidence provided from each modality. This enables the generation of claims that are highly related to the visual content or that require consideration of cross-modal evidence.



Figure 6: In this figure, we present additional claim examples from our dataset. While not all claims are entirely generated from the visual data, many can be verified by examining the visual content within the corresponding news cluster. This demonstrates that our dataset implicitly and explicitly contains multimodal claims, highlighting the importance of considering both textual and visual information for claim verification.

annotators are required to annotate them separately, as shown in Figure 12, ensuring that all AMR tuples coming from the AMR tree are labeled. For example, in Figure 12, the annotators need to go through a series of different AMR tuples for just one claim and label all elements inside the AMR image.

To ensure the quality and consistency of the human-labeled dataset, we provide the annotators with guidelines and examples for each label category. The annotators are also given the opportunity to discuss and resolve any disagreements or ambiguities in the labeling process. This collaborative approach helps to maintain a high level of inter-annotator agreement and reduces the potential for individual biases or errors. Our annotators consist of all the authors of this paper, each of whom is an expert in AMR and familiar with its properties and constraints. During the labeling process, the annotators are required to perform fine-grained labeling while adhering to AMR properties and constraints.

To ensure the quality and consistency of the fine-grained labels, we have established a set of guidelines that the annotators must follow:

- Adherence to AMR properties: The annotators must have a deep understanding of the properties and constraints of AMR, such as the semantic roles of nodes and the relationships between them. This knowledge is crucial for accurate, fine-grained labeling.

- Consistency with sample-level labels: The fine-grained labels should be consistent with the sample-level labels. For example, if the sample-level label is neutral, at least one AMR node must be labeled as neutral to maintain consistency.

- Maintenance of structural constraints: The annotators must ensure that the structural constraints within the AMR tree are preserved. This means that the labels of nodes and edges should be semantically consistent with each other. For instance, if a node is labeled as contradicted, the corresponding edge must also be labeled as contradicted to maintain the logical structure of the AMR.

- Collaboration and discussion: The annotators are encouraged to collaborate and discuss any ambiguities or disagreements in the labeling process. This collaborative approach helps to

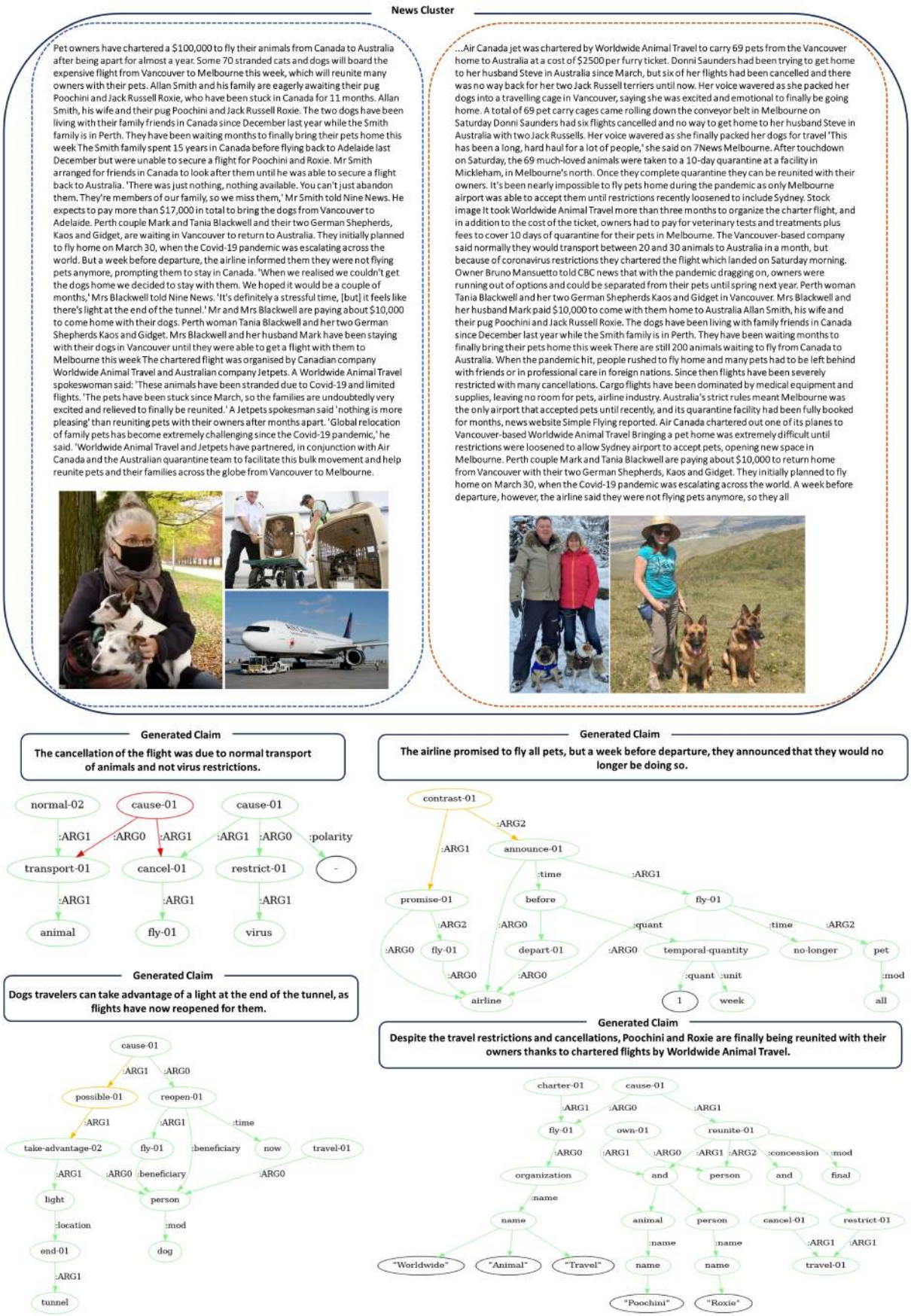


Figure 7: Results of the generated claims and the corresponding fine-grained level predictions. For instance, consider the generated claim in the top left corner. The ground truth label for this claim is "contradicted," as the flight cancellation was not caused by the normal transport of animals. Our model successfully detects this fact and assigns the correct fine-grained labels to the relevant parts of the claim.



Figure 8: We show a cluster of news documents containing multiple videos, images, and news articles. The cluster contains media about police officers pulling a man from a burning truck, along with cell phone video, body cam footage, and a press conference about the incident.

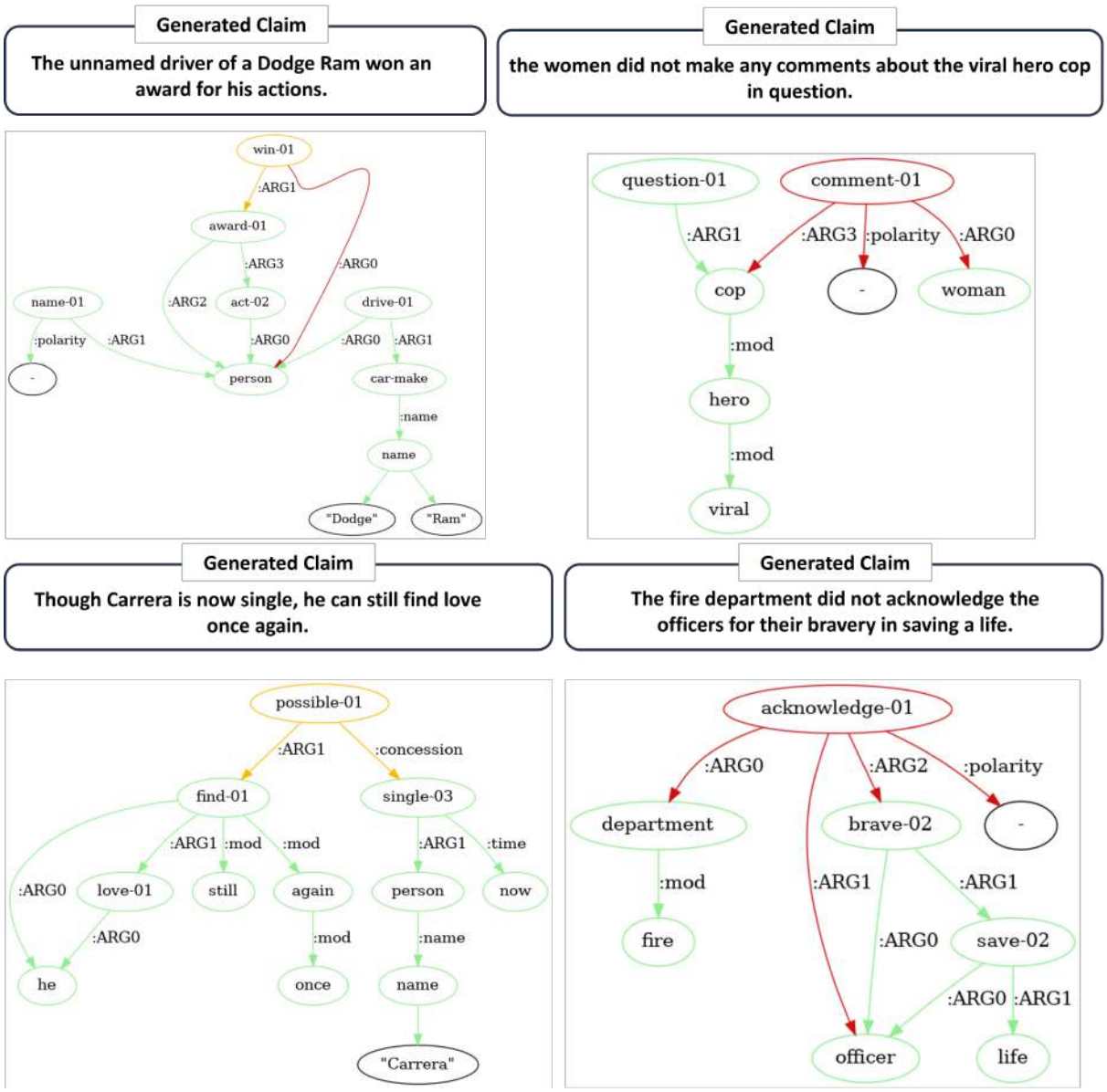


Figure 9: We show predictions of our model for a set of claims generated for the previous cluster.



Figure 10: This figure illustrates the labeling interface in LabelStudio, where annotators are required to review multiple news articles and their accompanying images before assigning labels to claims. This process can be time-consuming and challenging, as some claims rely on evidence scattered across small pieces of text or other modalities within the articles, demanding careful examination and synthesis of information from various sources to make accurate labeling decisions.

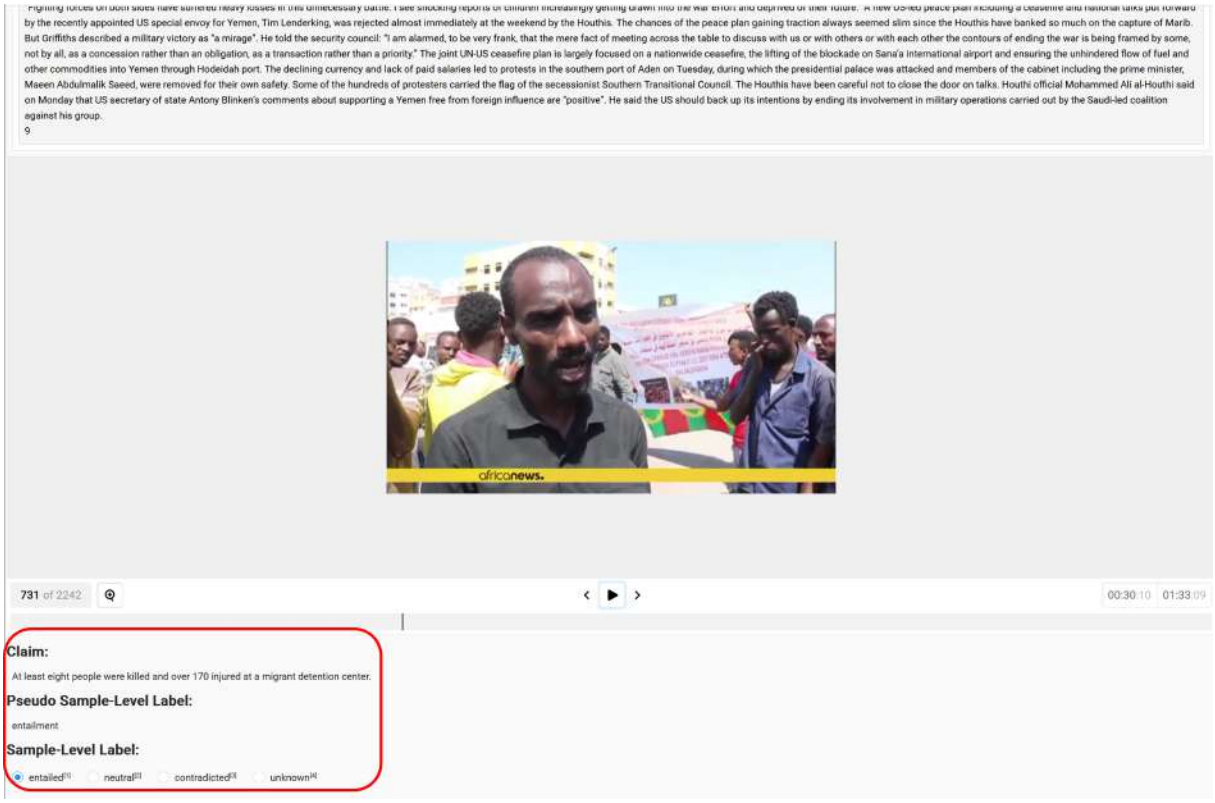


Figure 11: This figure shows the video examination process in the labeling interface, where annotators are tasked with reviewing videos associated with each news cluster, in addition to the news articles and corresponding images. The number of videos to be examined can range from none to a dozen per cluster with variable length. After thoroughly examining the evidence from news documents, images, and videos, the annotators are required to assign a logical label to the given claim, indicating its truthfulness based on the available multimodal information.

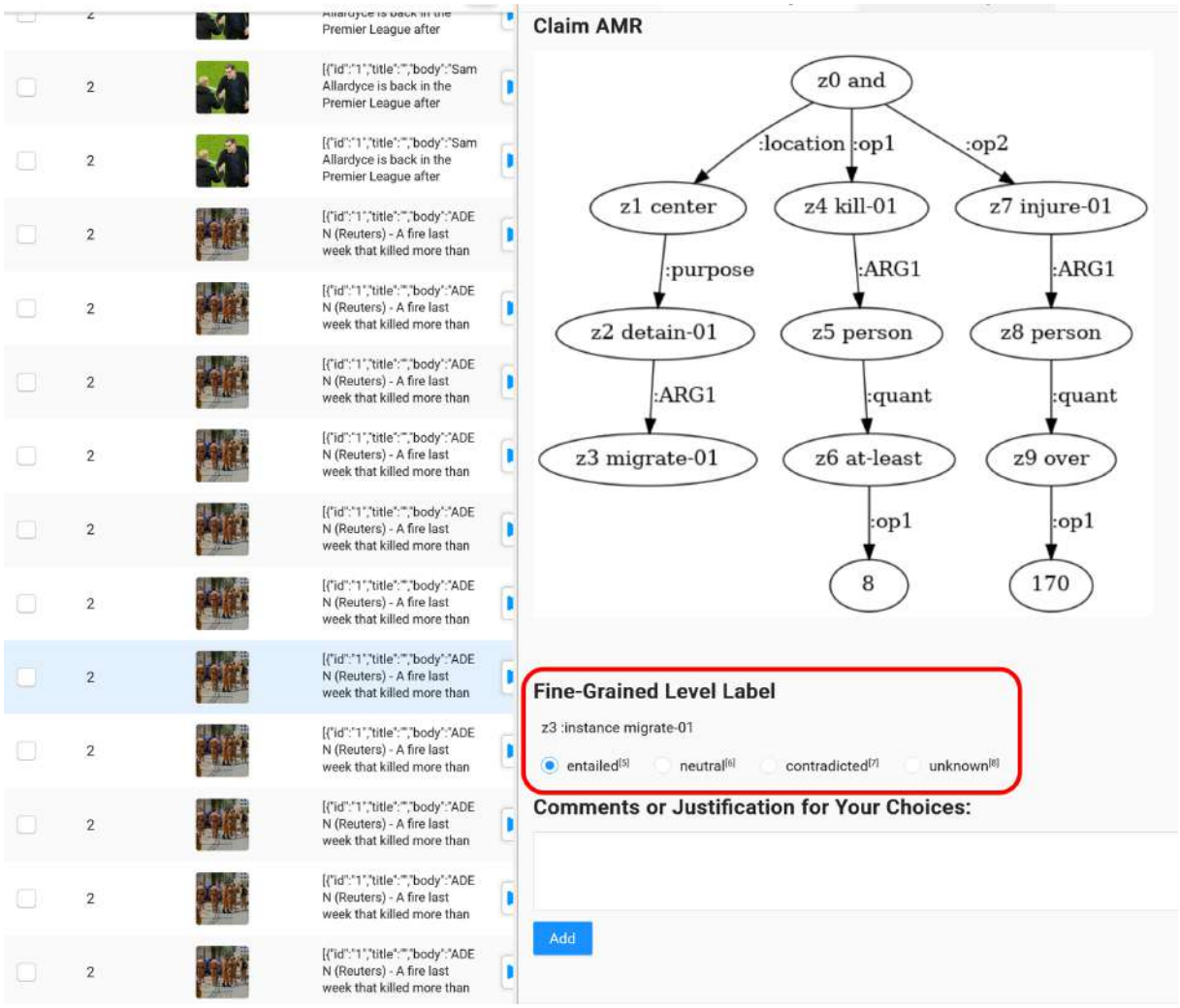


Figure 12: This figure depicts the fine-grained labeling process for AMR tuples in our dataset. Annotators are required to iterate through each AMR node and edge from the AMR tree, assigning fine-grained labels to evaluate the truthfulness of the claim at a more granular level. This process involves examining each tuple individually and making labeling decisions based on the available evidence from the news articles, images, and videos associated with the corresponding news cluster.

resolve any inconsistencies and ensures that the resulting labels are accurate and semantically consistent.

Adhering to these guidelines ensures that the fine-grained labels assigned by the annotators are semantically consistent within the AMR tree and accurately represent the information conveyed in the news clusters. However, when examining the IAA (IAA) scores for the human-labeled dataset, we observe a discrepancy between the sample-level and fine-grained level labels. The IAA for the sample-level labels is a high 93%, indicating strong agreement among the annotators when it comes to the overall veracity of the claims, suggesting that the annotators have a clear understanding of the broader context and are generally able to determine whether a claim is true, false, or neutral based on the available evidence. In contrast, the IAA for the fine-grained level labels is lower, at 68%, revealing that even when the annotators agree on the overall truthfulness of a claim, there can be disagreements when it comes to assigning labels to specific elements within the claim. This discrepancy highlights the complexity and nuance involved in fine-grained fact-checking, as different annotators may interpret the evidence differently or focus on different aspects of the claim when making their labeling decisions.

Table
reftab:misinfo_results presents the results of our model compared to the ground truth human labels, which are determined by the most voted label among the annotators. The results show that the entailment label accuracy of our model is close to human performance, indicating that the model can effectively identify claims that are supported by the available evidence. However, the model's performance on neutral and contradicted labels is not as high as its entailment accuracy, suggesting room for improvement in these areas. Despite this, the overall results demonstrate that our model can successfully assess the truthfulness of claims in this task, even though it may not yet match human performance across all label categories. The IAA scores further highlight the challenges associated with fine-grained fact-checking, even when all the experts involved in labeling the data do so with the utmost care and attention to detail.

A.4 LVLM Baselines

To evaluate the performance of LVLMs on fine-grained AMR prediction, we had to employ a workaround since these models do not natively support this task. Our approach involved using in-context learning to enable the LVLM models to perform fine-grained prediction at the word token level first. Once the models generated their predictions for the individual word tokens, we then mapped these results back to the corresponding nodes and edges in the AMR tree. This process allowed us to evaluate our dataset with LVLM models, even though they were not explicitly designed for this purpose.

We compare our model's performance with two state-of-the-art LVLMs trained on instructional data, which have demonstrated strong performance in tasks such as visual question answering and image captioning.

Our LVLM baselines include:

- **LLaVA** (Liu et al., 2023) is an instruction-tuned multimodal LVLM with strong image-text understanding capabilities. The model encodes image data using a pre-trained CLIP ViT-L/14 (Radford et al., 2021) and projects it into the Vicuna LLM's text embedding space (Chiang et al., 2023). It is tuned using large multimodal instructions curated via querying GPT-4 (Achiam et al., 2023).

- **MiniGPT-v2** (Chen et al., 2023) is an improved version of MiniGPT-4 (Zhu et al., 2023) and has a simpler architecture. It uses EVA (Fang et al., 2023) as the pretrained CLIP image encoder and LLaMA-2-Chat (Touvron et al., 2023) as the LLM backbone. The model demonstrates strong performance in multimodal understanding on numerous image-text tasks.

Prompts We obtain both sample-level and fine-grained predictions from the LVLM baselines by prompting them in a zero-shot manner. In the single-document setting (*i.e.*, MOCHEG), we provide the LVLM with multimodal evidence – including a text document and its corresponding image – alongside a claim and an instructional question. Given the evidence, the prompt instructs the model to verify either the entire claim or words within the claim, corresponding with the sample-level task or the fine-grained task, respectively. Table **??** shows an example of a text prompt constructed from an

example in the MOCHEG dataset, and Table ?? includes all the different questions to be prompted for that example. In the multi-document setting (*i.e.*, M^3DC), we carry out the same steps for each document in a document group. We then perform majority voting among the group's predictions to compute the final prediction.

A.5 Limitations and Future Works

While our proposed approach for constructing a fact-checking dataset with fine-grained labels integrates multimodal and multi-document data, there are still several limitations that need to be addressed in future research. One of the main limitations is that the visual evidence in our dataset consists of grounding captions generated from images and video frames, resulting in a heavy reliance on textual data rather than plain visual data. Additionally, we do not include video transcripts or audio information in the KG, further emphasizing the textualism of our dataset. Given the nature of our dataset, which primarily consists of news documents where textual evidence dominates over other modalities, it's expected that the constructed dataset and the resulting model focus more on textual input, including the generated claims and information needed for reasoning.

Another limitation is that our model relies on the underlying assumption that genuine news articles are consistent, trustworthy, and complementary. However, there is a possibility that articles from the same news cluster can contain inconsistent information. For example, one article could report that there were nine people at the scene, while an image in another article only shows seven people. Moreover, certain types of human-written fake news documents, such as conspiracy theories, tend to be closely related and convey highly similar information due to shared biases or the intent to manipulate readers in a specific way. These issues of inconsistent information and similarity among fake news articles may limit the performance of our proposed system when applied to real-world data.

To address these limitations, future work could focus on the following areas: (1) incorporating more diverse modalities, such as raw visual and audio data, into the KG and the resulting dataset to reduce the reliance on textual data; (2) integrating commonsense reasoning techniques into the model to better capture complex contradictions and improve the system's ability to identify inconsistency and misinformation; (3) exploring alternative approaches that do not rely solely on the assumption of consistency among genuine news articles, thus improving the system's robustness when dealing with real-world fake news.

By addressing these limitations and exploring new research directions, we aim to enhance the performance and applicability of our proposed model in real-world scenarios, ultimately contributing to the fight against the spread of misinformation. We plan to publicly release our multimodal, multi-document dataset and the proposed model implementation in the future to foster further research in this area.

A.6 Ethical Considerations

In this work, our primary objective is to advance the state-of-the-art in fact-checking by analyzing multiple multimedia documents on the same topic. To achieve this goal, we have constructed a new benchmark dataset using the proposed methodology and developed a detector capable of determining the truthfulness of a given claim. To facilitate future research and benefit the community, we will make the constructed dataset and detector codes available, serving as a valuable reference for researchers and practitioners in the field.

However, we acknowledge that our work, like any research involving text generation, carries the risk of being misused to produce false information with the intent to mislead or manipulate readers. We want to clarify that the dataset and model we constructed do not contain true claims, but rather claims generated from models. The dataset and model are intended solely for research purposes and should not be used to suppress opinions or make misjudgments. We strongly emphasize the importance of responsible and ethical use of these resources in the pursuit of advancing fact-checking techniques.

Given the evidence (including the image and a text article) and a text claim, please indicate whether **a word in the claim** is supported or refuted by the evidence.
r0.3



Article: A photograph purportedly showing a moose and two calves enjoying a kiddie pool as they watched a car burn across the street has been circulating online for several years. While it is frequently shared as a genuine (albeit bizarre) item, this image is a composite of at least two separate photographs. The photograph of the car on fire first appeared online when it was published on Reddit in May 2013. It seems as if they were trying to jump-start it. Obviously, they don't know their cars too well. The whole neighborhood has gathered for the impromptu neighborhood bonfire. While we haven't been able to locate the specific origin of the moose image, we know that the photograph was also posted separately to Reddit in May 2013: Unsurprisingly, the first version of the image featuring moose in a kiddie pool watching a car fire appeared on (of course) Reddit, shortly after the two source images were posted.

Claim: A photograph shows a moose enjoying a wading pool while watching a car burn.
Question: Is the word "moose" in the claim true, false, or neutral with regard to the evidence?
Output "True" if the evidence supports the word, "False" if the evidence contradicts the word, or "Neutral" if it is neither supported nor refuted.
Answer:

Table 5: An example of a zero-shot prompt to be fed into LVLMs for sample-level and fine-grained predictions, constructed from a data example in the MOCHEG dataset. If the model

Sample-level question: Is the claim true, false, or neutral with regards to the evidence?
Fine-grained questions:

- Is the word "photograph" in the claim true, false, or neutral with regards to the evidence?
- Is the word "shows" in the claim true, false, or neutral with regards to the evidence?
- Is the word "moose" in the claim true, false, or neutral with regards to the evidence?
- Is the word "enjoying" in the claim true, false, or neutral with regards to the evidence?
- Is the word "wading" in the claim true, false, or neutral with regards to the evidence?
- Is the word "pool" in the claim true, false, or neutral with regards to the evidence?
- Is the word "watching" in the claim true, false, or neutral with regards to the evidence?

Table 6: Example questions to prompt sample-level and fine-grained zero-shot predictions. We only construct fine-grained questions on words that can be mapped to AMR triple annotations to ensure ground truths for evaluation.